

DSD Final Project Scores

1. BrPred

(1) Total execution cycles of given I_mem_BrPred:

截圖:

```
[INFO]: Testing with pattern <BrPred>
-----

START!!! Simulation Start .....

-----

=====
Reset ...
=====
sh: ../00_TESTBED/info/success: Permission denied
=====

\\(^o^)/ CONGRATULATIONS!! The simulation result is PASS!!!

=====
$finish called from file "../00_TESTBED/testbench/Final_tb.v", line 269.
$finish at simulation time 1579033
```

(2) Total execution cycles of given I_mem_hasHazard:

截圖:

```
[INFO]: Testing with pattern <hasHazard>
-----

START!!! Simulation Start .....

-----

=====
Reset ...
=====
sh: ../00_TESTBED/info/success: Permission denied
=====

\\(^o^)/ CONGRATULATIONS!! The simulation result is PASS!!!

=====
$finish called from file "../00_TESTBED/testbench/Final_tb.v", line 269.
$finish at simulation time 6706033
```

(3) Clock cycle for post-syn simulation (cycle in testbench, not cycle in sdc): (ns)

3ns

2. Compressed instructions

(1) Total Simulation Time of given I_mem_compression: (ns)

截圖:

```

[INFO]: Testing with pattern <compression>
-----

START!!! Simulation Start .....

-----

=====
Reset ...
=====
sh: ../00_TESTBED/info/success: Permission denied
=====

\\(^o^)/ CONGRATULATIONS!! The simulation result is PASS!!!

=====
$finish called from file "../00_TESTBED/testbench/Final_tb.v", line 269.
$finish at simulation time          1990033

```

(2) Clock cycle for post-syn simulation (cycle in testbench, not cycle in sdc): (ns)

3ns

3. Multiplication instructions

(1) Total Simulation Time of given Mul/I_mem: (ns)

截圖:

```

[INFO]: Testing with pattern <Mul>
-----

START!!! Simulation Start .....

-----

=====
Reset ...
=====
sh: ../00_TESTBED/info/success: Permission denied
=====

\\(^o^)/ CONGRATULATIONS!! The simulation result is PASS!!!

=====
$finish called from file "../00_TESTBED/testbench/Final_tb.v", line 269.
$finish at simulation time          718033

```

(2) Clock cycle for post-syn simulation (cycle in testbench, not cycle in sdc): (ns)

3ns

4. Q_sort & Conv & LFSR_HIST

(1) Area: (483059um²)

截圖:

Combinational area:	278998.244294
Buf/Inv area:	38512.308615
Noncombinational area:	204061.428898
Macro/Black Box area:	0.000000
Net Interconnect area:	3825286.257721
 Total cell area:	 483059.673192
Total area:	4308345.930912

(2) Total Simulation Time of Q_Sort: (316447ns)

(either using compressed or uncompressed instructions)

截圖:

```
[INFO]: Testing with pattern <QSort_uncompressed>
-----

START!!! Simulation Start .....

-----

=====
Reset ...
=====
sh: ../00_TESTBED/info/success: Permission denied
=====

\\(^o^)/ CONGRATULATIONS!! The simulation result is PASS!!!

=====
$finish called from file "../00_TESTBED/testbench/Final_tb.v", line 269.
$finish at simulation time 316447033
```

(3) Total Simulation Time of Conv: (58963ns)

(either using compressed or uncompressed instructions)

截圖:

```
[INFO]: Testing with pattern <Conv_uncompressed>
-----

START!!! Simulation Start .....

-----

=====
Reset ...
=====
sh: ../00_TESTBED/info/success: Permission denied
=====

\\(^o^)/ CONGRATULATIONS!! The simulation result is PASS!!!

=====
$finish called from file "../00_TESTBED/testbench/Final_tb.v", line 269.
$finish at simulation time 58963033
```

(4) Total Simulation Time of LFSR_HIST: (3250843ns)

截圖:

```

[INFO]: Testing with pattern <LFSR_HIST>
-----
START!!! Simulation Start .....
-----

=====
Reset ...
=====
sh: ../00_TESTBED/info/success: Permission denied
=====

\(^o^)/ CONGRATULATIONS!! The simulation result is PASS!!!

=====
$finish called from file "../00_TESTBED/testbench/Final_tb.v", line 269.
$finish at simulation time          3250843033

```

(5) $A \times T_{Q_sort} \times T_{Conv} \times T_{LFSR_HIST} : (um^2 * ns^3)$

2.93 e22

(6) Clock cycle for post-syn simulation (cycle in testbench, not cycle in sdc): (ns)

3ns